



SCOPE OF ACCREDITATION TO ISO/IEC 17025:2005
& ANSI/NCSL Z540-1-1994

ALLTITE, INC.
 1600 Murdock,
 Wichita, KS 67214
 Gregory Dunsmoor Phone: 316 686 3010

CALIBRATION

Valid To: September 30, 2018

Certificate Number: 3040.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Dimensional

Parameter/Equipment	Range	CMC ² (±)	Comments
Calipers ³ – Dial, Digital, Vernier			
500 µin Graduation 1000 µin Graduation	Up to 60 in	570 µin + 9 µin/in 560 µin + 13 µin/in	Gage blocks
Micrometers ³			
Outside –			
50 µin Graduation 100 µin Graduation 1000 µin Graduation	Up to 48 in	47 µin + 8.1 µin/in 93 µin + 8.6 µin/in 570 µin + 12 µin/in	Gage blocks
Inside –			
1000 µin Graduation	Up to 40 in	600 µin + 3 µin/in 50 µin + 10 µin/in	Gage blocks ULM
Depth –			
1000 µin Graduation	Up to 12 in	580 µin + 0.88 µin/in	Gage blocks w/ surface plate

John C. Burt

Parameter/Equipment	Range	CMC ² (±)	Comments
Micrometers ³ (Cont.) – 3 Point Bore – 50 µin Graduation 200 µin Graduation Bench – 10 µin Graduation	Up to 3 in Up to 2 in	93 µin + 7.1 µin/in 190 µin + 3.9 µin/in 22µin	Setting rings Gage blocks
Indicators – 50 µin Graduation 100 µin Graduation 500 µin Graduation 1000 µin Graduation	Up to 4 in	49 µin + 5.5 µin/in 69 µin + 4.3 µin/in 290 µin + 1.8 µin/in 580 µin + 0.6 µin/in	ULM
Pin Gages ³	Up to 1 in	32 µin	Bench micrometer
Cylindrical Rings	Up to 10 in	20 µin + 8.6 µin/in	ULM w/master ring
Height Gage ³ – 10 µin Graduation 100 µin Graduation 1000 µin Graduation	Up to 48 in	170 µin + 5.9 µin/in 180 µin + 6.0 µin/in 600 µin + 2.8 µin/in	Gage blocks
Length Standards	Up to 32 in	17 µin + 11 µin/in	ULM

II. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 4} (±)	Comments
DC Voltage – Generate ³	(0 to 320) mV (0.32 to 3.2) V (3.2 to 32) V (32 to 320) V (320 to 1050) V	20 µV 0.0019 V 0.0021 V 0.020 V 0.065 V	Fluke 5502A
DC Current – Generate ³	(0 to 320) µA 320 µA to 3.2 mA (3.2 to 32) mA (32 to 320) mA (0.32 to 1.0) A (1.0 to 3.0) A (3.0 to 11) A (11 to 20) A (0 to 500) A (500 to 1000) A	0.069 µA 0.057 µA 0.058 mA 0.036 mA 0.42 mA 1.2 mA 6.5 mA 25 mA 2.1 A 4.6 A	Fluke 5502A w/ 50 turn coil
Resistance – Generate ³	(0 to 10.999) Ω (11 to 33) Ω (33 to 110) Ω (110 to 330) Ω (330 to 1100) Ω (1.1 to 3.3) kΩ (3.3 to 11) kΩ (11 to 33) kΩ (33 to 110) kΩ (110 to 330) kΩ (330 to 1100) kΩ (1.1 to 3.3) MΩ (3.3 to 11) MΩ (11 to 33) MΩ (110 to 330) MΩ (330 to 1100) MΩ	11 mΩ 19 mΩ 25 mΩ 0.047 Ω 0.12 Ω 0.47 Ω 20 Ω 3.7 Ω 13 Ω 49 Ω 170 Ω 0.6 kΩ 6.8 kΩ 33 kΩ 1.6 MΩ 17 MΩ	Fluke 5502A



Parameter/Range	Frequency	CMC ^{2,4} ($\pm$)	Comments
AC Voltage – Generate ³			
(0 to 33) mV _{AC}	(10 to 45) Hz (45 to 10 000) Hz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.046 mV 0.05 mV 0.074 mV 0.086 mV 0.16 mV 1.7 mV	Fluke 5502A
(33 to 330) mV _{AC}	(10 to 45) Hz (45 to 10 000) Hz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	0.19 mV 0.11 mV 0.23 mV 0.34 mV 0.86 mV 1.8 mV	
(330 to 3300) mV _{AC}	(10 to 45) Hz (45 to 10 000) Hz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 500) kHz	1.6 mV 1.1 mV 2.2 mV 3.1 mV 7.1 mV 16 mV	
3.3 to 33 V _{AC}	(10 to 45) Hz (45 to 10 000) Hz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	12 mV 9.6 mV 22 mV 31 mV 71 mV	
33 V to 330 V _{AC}	(45 to 1000) Hz (1 to 10) kHz (10 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.18 V 0.25 V 0.28 V 0.37 V 0.56 V	
330 V to 1020 V _{AC}	45 Hz to 1000 kHz (1 to 5) kHz (5 to 10) kHz	0.52 V 0.82 V 0.94 V	



Parameter/Range	Frequency	CMC ^{2, 4} ($\pm$)	Comments
AC Current – Generate ³			
(29 to 330) μA_{AC}	(10 to 20) Hz (20 to 45) Hz (45 to 1000) Hz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.76 μA 0.59 μA 0.51 μA 1.1 μA 2.8 μA 5.7 μA	Fluke 5502A
(0.33 to 3.29) mA _{AC}	(10 to 20) Hz (20 to 45) Hz (45 to 1000) Hz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	6.8 μA 3.5 μA 4.6 μA 6.8 μA 17 μA 35 μA	
(3.3 to 3.29) mA _{AC}	(10 to 20) Hz (20 to 45) Hz (45 to 1000) Hz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	61 μA 32 μA 16 μA 29 μA 69 μA 0.14 mA	
(33 to 329) mA _{AC}	(10 to 20) Hz (20 to 45) Hz (45 to 1000) Hz (1 to 5) kHz (5 to 10) kHz (10 to 30) kHz	0.1 mA 0.34 mA 0.2 mA 0.43 mA 0.78 mA 1.5 mA	
(0.33 to 1.09)	(10 to 45) Hz (45 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	3 mA 0.65 mA 7.5 mA 32 mA	
(1.1 to 2.99) A _{AC}	(10 to 45) Hz (45 to 1000) Hz (1 to 5) kHz (5 to 10) kHz	5.6 mA 2.1 mA 19 mA 80 mA	
(3 to 10.9) A _{AC}	(10 to 45) Hz (45 to 1000) Hz (1 to 5) kHz	8.9 mA 13 mA 330 mA	
(11 to 20.5) A _{AC}	(10 to 45) Hz (45 to 1000) Hz (1 to 5) kHz	29 mA 35 mA 610 mA	
(0 to 150) A _{AC}	(10 to 100) Hz	7.6 mA	
(150 to 1000) A _{AC}	(10 to 100) Hz	2.1 mA	w/ 50 turn coil



III. Mechanical

Parameter/Equipment	Range	CMC ^{2, 4} ($\pm$)	Comments
Torque –			
Manual Torque Wrenches ³	(2 to 24) in·lbf	0.19 in·lbf + 0.023 in·lbf/ in·lbf	ISO 6789:2003 Manually applied and measured with a transducer
	(20 to 24) in·lbf	1.5 in + 0.012 in·lbf/ in·lbf	
	(20 to 200) ft·lbf	0.53 ft·lbf + 0.0089 ft·lbf/ft·lbf	
	(200 to 2000) ft·lbf	2.4 ft·lbf + 0.0075 ft·lbf/ft·lbf	
Hydraulic Torque Wrenches ³	(500 to 5000) ft·lbf	5.2 ft·lbf + 0.0014 ft·lbf/ft·lbf	Hydraulic pressure measured and compared to torque measured by transducers
	(2000 to 20 000) ft·lbf	8.6 ft·lbf + 0.0076 ft·lbf/ft·lbf	
	(20 001 to 40 000) ft·lbf	43 ft·lbf + 0.0091 ft·lbf/ft·lbf	
Pneumatic Torque Gun ³	(0 to 5000) ft·lbf	2.7 ft·lbf + 0.0016 ft·lbf/ft·lbf	Pneumatic pressure set and torque measured with transducer
Electronic Torque Gun ³	(0 to 5000) ft·lbf	1.6 ft·lbf + 0.0024 ft·lbf/ft·lbf	Digital torque set and torque measured with transducer
Torque Multiplier ³	(0 to 20000) ft·lbf f=(output)	6.4 ft·lbf + 0.017 ft·lbf/ft·lbf	Input and output torque measured with torque transducers
Torque Transducers	(2 to 24) in·lbf	0.011 in·lbf + 0.0018 in·lbf/ in·lbf	Torque arm with weights
	(24 to 240) in·lbf	0.14 in·lbf + 0.00022 in·lbf/ in·lbf	
	(20 to 200) ft·lbf	0.059 ft·lbf + 0.00013 ft·lbf/ ft·lbf	
	(200 to 2000) ft·lbf	0.65 ft·lbf + 0.00014 ft·lbf/ft·lbf	
	(2000 to 20 000) ft·lbf	4.6 ft·lbf + 0.00015 ft·lbf/ ft·lbf	



Parameter/Equipment	Range	CMC ² (±)	Comments
Pressure ³ – Hydraulic Gauges	(0 to 300) psig (300 to 3000) psig (3001 to 10 000) psig (10 000 to 30 000) psig	0.19 psig + 44 µpsig/psig 1.3 psig + 9.6 µpsig/psig 4.8 psig + 32 µpsig/psig 60 psig + 0.93 µpsig/psig	Comparison with standard hydraulic pressure gage

IV. Time & Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Infrared Thermometer (Non-Contact)	35° C to 500 °C	1.3° C + 0.25 % of rdg	Fluke 4181 blackbody

¹ This laboratory offers commercial calibration service and field calibration service.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA R104 – *General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMC's are expressed as either a specific value that covers the full range or as a percent or fraction of the reading plus a fixed floor specification.





Accredited Laboratory

A2LA has accredited

ALLTITE, INC

Wichita, KS

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994 and R205 – Specific Requirements: Calibration Laboratory Accreditation Program. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (*refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009*).



Presented this 27th day of October 2016.

A handwritten signature in blue ink, appearing to read 'J. C. Bunt'.

Senior Director of Quality and Communications
For the Accreditation Council
Certificate Number 3040.01
Valid to September 30, 2018

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.